

Discrete Structures

Quiz 2 (Section H, F)

DATE

Q1: Write Negation for each statement

i) $\forall$ real number x , if $x^2 \geq 1$ then $x > 0$

Let $p: x^2 \geq 1$, $q: x > 0$

$\forall x \in \mathbb{R}, p \rightarrow q$

Negation:

$\exists x \in \mathbb{R}, p \wedge \neg q$

There exists a real number x such that $x^2 \geq 1$ and $x \leq 0$.

ii) $\forall$ integer d , if $6/d$ is an integer then $d=3$

Let $p: 6/d$ is an integer, $q: d=3$

$\forall d \in \mathbb{Z}, p \rightarrow q$

Negation:

$\exists d \in \mathbb{Z}, p \wedge \neg q$

There exists an integer d such that $6/d$ is an integer and $d \neq 3$

iii) $\forall x \in \mathbb{R}$, if $x(x+1) > 0$ then $x > 0$ or

Let $p: x(x+1) > 0$, $q: x > 0$, $r: x < -1$, $s: x < -1$

$\forall x \in \mathbb{R}, p \rightarrow (q \vee r)$

Negation:

$\exists x \in \mathbb{R}, p \wedge \neg q \wedge \neg r$

There exists a real number x , such that $x(x+1) > 0$ and $x \leq 0$ and $x \geq -1$

iv) $\forall n \in \mathbb{Z}$, if n is prime then n is odd or $n=2$.

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Let $p: n$ is prime, $q: n$ is odd, $r: n=2$
 $\forall n \in \mathbb{Z}, p \rightarrow (q \vee r)$

Negation:

$\exists n \in \mathbb{Z}, p \wedge \neg q \wedge \neg r$

There exist an integer n , such that
 n is prime and n is even and
 $n \neq 2$.

v) $\forall$ integers a, b, c , if $a-b$ is even
and $b-c$ is even, then $a-c$ is even.

Let $p: a-b$ is even

$q: b-c$ is even

$r: a-c$ is even

$\forall$ integers $a, b, c, (p \wedge q) \rightarrow r$

Negation:

$\exists a, b, c \in \mathbb{Z}, p \wedge q \wedge \neg r$

There exist integers a, b, c such that
 $a-b$ is even and $b-c$ is even and
 $a-c$ is odd.

vi) $\forall$ integers n , if n is divisible by 6
then n is divisible by 2 and
 n is divisible by 3.

Let $p: n$ is divisible by 6

$q: n$ is divisible by 2

$r: n$ is divisible by 3

$\forall n \in \mathbb{Z}, p \rightarrow (q \wedge r)$

Negation:

$\exists n \in \mathbb{Z}, p \wedge \neg q \vee \neg r$

There exist integers n , such that n is
divisible by 6 and n is not divisible by
2 or n is not divisible by 3.

vii) If the square of integers is odd, then the integer is odd.

Let p : Square of integer is odd.

q : integer is odd.

$\forall x \in \mathbb{Z}, p \rightarrow q$

Negation:

$\exists x \in \mathbb{Z}, p \wedge \neg q$

There exists an integer x such that square of x is odd and x is even.

Q2: $\exists x \in \mathbb{R}$ such that $x^2 = 2$.

Which of the following are equivalent ways of expressing this statement.

a) The square of each real number is 2
No

b) Some real numbers have square 2
Yes

c) The number x has square, for some real number x .
Yes

d) If x is a real number, then $x^2 = 2$
No

e) Some real number has square 2
Yes

f) There is at least one real number whose square is 2.
Yes.